

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination – 2025

Course: B.Tech.

Branch : Artificial Intelligence (AI) and Data Science

Semester : V

Subject Code & Name: BTAIPE504C_SENSORS & ROBOTICS TECHNOLOGY

Max Marks: 60

Date: 21/02/2026

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	Marks
Q. 1 Objective type questions. (Compulsory Question)	12
1 Which of the following is a systematic error?	1
a) Gross error b) Instrumental error c) Random fluctuation d) Human mistake	
2 Sensitivity of a sensor refers to	1
a) Speed of response b) Output change per unit input c) Accuracy d) Drift	
3 Which is a passive sensor?	1
a) Thermocouple b) Piezoelectric sensor c) RTD d) Solar cell	
4 Amplifier in DAQ system is used to	1
a) Reduce noise b) Increase signal strength	
c) Store data d) Convert analog to digital	
5 Servo motors are mainly used where	1
a) Speed control is not required b) Precise position control is required	
c) No feedback is used d) Only AC supply is available	
6 A 3/2 valve has	1
a) 3 ports, 2 positions b) 2 ports, 3 positions	
c) 3 ports, 3 positions d) 2 ports, 2 positions	
7 Recognition subsystem of robot includes	1
a) Actuators b) Sensors c) Motors d) Power supply	
8 Number of independent movements of robot is called	1
a) Stroke b) Reach c) Axis d) Payload	
9 Forward kinematics is used to determine	1
a) Joint angles from position b) End-effector position from joint angles	
c) Speed d) Torque	
10 Jacobian matrix relates	1
a) Position and velocity b) Voltage and current	
c) Torque and speed only d) Force and temperature	
11 Vacuum grippers are mainly used for handling	1
a) Rough surfaces b) Heavy metals c) Smooth flat objects d) Liquids	
12 A-star algorithm is an example of	1
a) Heuristic search b) Sorting algorithm	
c) Compression method d) Encryption method	

Q. 2 Solve the following.	12
A) Explain static and dynamic characteristics of transducers in detail with suitable examples.	6
B) Explain vision sensing using digital cameras. Discuss image acquisition, resolution, frame rate, and challenges in robotic vision systems.	6
Q.3 Solve the following.	12
A) Compare single-channel and multi-channel data acquisition systems. Discuss their applications in aerospace and manufacturing industries.	6
B) Describe the construction and working of stepper motor and servo motor. Compare their control methods and industrial applications.	6
Q. 4 Solve Any Two of the following.	12
A) Explain various robot specifications such as number of axes, reach, stroke, payload capacity, repeatability and accuracy.	6
B) Classify robots based on drive technology, work envelope geometry, and motion control methods. Compare different types with examples.	6
C) Discuss safety measures in robotics. Explain hazards associated with industrial robots and methods to ensure human–robot collaboration safety.	6
Q.5 Solve Any Two of the following.	12
A) Explain forward and inverse kinematics of a robotic manipulator. Discuss challenges in solving inverse kinematics.	6
B) Describe Denavit–Hartenberg (DH) parameters and their role in coordinate transformation of robotic arms.	6
C) A 2-link planar robotic arm has link lengths $L_1 = 40$ cm and $L_2 = 30$ cm. The joint angles are $\theta_1 = 30^\circ$ and $\theta_2 = 45^\circ$. Calculate: a) The position (X, Y) of the end-effector. b) The distance of the end-effector from the origin.	6
Q. 6 Solve Any Two of the following.	12
A) For a 2-DOF planar robot with $L_1 = 1$ m and $L_2 = 0.8$ m, the joint angles are $\theta_1 = 45^\circ$ and $\theta_2 = 30^\circ$. The joint angular velocities are $\omega_1 = 0.5$ rad/s and $\omega_2 = 0.3$ rad/s. Calculate: a) The Jacobian matrix. b) The linear velocity of the end-effector.	6
B) Explain different types of grippers. Discuss their applications.	6
C) Discuss perception, localization, and mapping in mobile robotics. Explain the concept of probabilistic robotics.	6

*** End ***